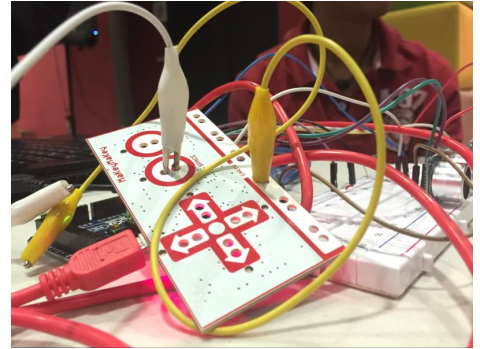


Makey Makey STEM Kit

WHAT'S IN THE KIT

- 16 Makey Makey boards
- Alligator clips
- USB cables



QUICK-START GUIDE

1. Plug the Makey Makey into a computer via USB — no software needed.
2. Clip one alligator clip to 'Earth' and hold the other end.
3. Clip another to an arrow key and touch anything conductive to trigger it.
4. Pair with Scratch to build custom instruments and controllers.

FULL LESSONS IN THIS GUIDE

GRADES K-2	Fruit Piano	page 2
GRADES 3-5	Interactive Poster	page 3
GRADES 6-8	Game Controller Design	page 4

PRO TIP: Playdough is the most reliable conductor for little hands — keep a tub handy.

Makey Makey STEM Kit

LESSON · GRADES K-2

Fruit Piano

LEARNING OUTCOMES

- Students will sort materials by whether they conduct electricity
- Students will explain that a circuit must be complete to work
- Students will make a prediction and test it

MATERIALS

- Makey Makey boards + clips (1 per small group)
- Test objects: fruit, playdough, coins, erasers, paper, foil
- Device per group with a Scratch piano project loaded
- Prediction sort mat (works / doesn't work)

INSTRUCTIONS

1. Hook one banana to a piano key and play it — instant attention.
2. Show the two-hand rule: one hand on Earth clip, one on the object.
3. Groups predict-sort their objects on the mat before testing.
4. Test each object; move it to the correct side after the evidence.
5. Circle up: what do all the 'works' objects have in common?
6. Finish with a free-play fruit piano jam.

ACTIVITY

Fruit Piano — predict, test, and sort conductors from insulators, then play music through a banana. Playdough controllers are the reliable crowd-pleaser for little hands.

Makey Makey STEM Kit

LESSON · GRADES 3-5

Interactive Poster

LEARNING OUTCOMES

- Students will design a physical interface with conductive touchpoints
- Students will record and trigger audio in Scratch
- Students will combine research content with an interactive presentation

MATERIALS

- Makey Makey boards + clips
- Poster board, foil, brads, glue
- Devices with Scratch (recording enabled)
- Research notes from a current unit topic

INSTRUCTIONS

1. Show a finished interactive poster example first.
2. Students plan: 4-6 touchpoints, each triggering a recorded fact.
3. Record narrations in Scratch (quiet corner or Sound Shield station).
4. Build posters with foil touchpoints wired to Makey Makey inputs.
5. Test every touchpoint; troubleshoot connections with the two-hand rule.
6. Museum walk: half present at their posters, half tour, then swap.

ACTIVITY

Interactive Poster — a research poster (habitats, biographies, book reports) where touching foil icons plays student-recorded narration. Instantly museum-grade.

Makey Makey STEM Kit

LESSON · GRADES 6-8

Game Controller Design

LEARNING OUTCOMES

- Students will design a custom game controller for usability
- Students will prototype with conductive materials and iterate from user testing
- Students will explain the circuit path in their controller

MATERIALS

- Makey Makey boards + clips
- Cardboard, foil, copper tape, playdough
- Devices with a Scratch game (their own or provided)
- User-test feedback form

INSTRUCTIONS

1. Frame the design brief: build a controller that fits the game — a steering wheel, dance pad, drum set.
2. Sketch first: where do hands go? Where is Earth? Which inputs map where?
3. Prototype with cardboard and foil; wire to arrow keys/space.
4. User testing: another team plays for 3 minutes and completes the form.
5. Iterate on the top complaint (usually grounding or button spacing).
6. Demo day: 60-second pitch on their design decisions.

ACTIVITY

Game Controller Design — UX meets circuitry. The user-testing round is where students discover a controller that makes sense to its builder can baffle everyone else.